KAIZEN MATHS

Lesson starter

Modulus and Absolute Value

Inequalities

Name _____

S1

Solve: $|x| < 9$

S2

Solve: $|2x| + 7 \geq 19$

S3

Solve: $|2x + 1| \geq 16$

S4

Solve: $|4x - 3| + 4 > 15$

Show your working

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ALG-C00B-L3 · S

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Exit ticket

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E1

Solve: $|3x| < 9$

E2

Solve: $|2x| - 3 > 5$

E3

Solve: $|5x + 6| < 12$

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ALG-C00B-L3 · E

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Teacher answers & working

Modulus and Absolute Value · Inequalities

ALG-C00B-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Solve: $|x| < 9$

Answer:

$-9 < x < 9$

Working:

Convert to a compound inequality

$-9 < x < 9$

Solve both boundary equations

Lower boundary

$x = -9$

$x = -9$

Upper boundary

$x = 9$

$x = 9$

Write the values between the boundaries

$-9 < x < 9$

Number line: (-9, 9)

$-9 < x < 9$

Teacher answers & working

Modulus and Absolute Value · Inequalities

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S2

Solve: $|2x| + 7 \geq 19$

Answer:

$x \leq -6$ or $x \geq 6$

Working:

Isolate the absolute value

$|2x| \geq 19 - 7 = 12$

Split into two outside cases

$2x \leq -12$ or $2x \geq 12$

Solve each case

Left side

$2x \leq -12$

$x \leq -6$

Right side

$2x \geq 12$

$x \geq 6$

Write the outside intervals

$x \leq -6$ or $x \geq 6$

Number line: $(-\infty, -6] \cup [6, \infty)$

$x \leq -6$ or $x \geq 6$

Teacher answers & working

Modulus and Absolute Value · Inequalities

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S3

Solve: $|2x + 1| \geq 16$

Answer:

$x \leq -8.5$ or $x \geq 7.5$

Working:

Split into two outside cases

$2x + 1 \leq -16$ or $2x + 1 \geq 16$

Solve each case

Left side

$2x + 1 \leq -16$

$x \leq -8.5$

Right side

$2x + 1 \geq 16$

$x \geq 7.5$

Write the outside intervals

$x \leq -8.5$ or $x \geq 7.5$

Number line: $(-\infty, -8.5] \cup [7.5, \infty)$

$x \leq -8.5$ or $x \geq 7.5$

Teacher answers & working

Modulus and Absolute Value · Inequalities

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S4

Solve: $|4x - 3| + 4 > 15$

Answer:

$x < -2$ or $x > 3.5$

Working:

Isolate the absolute value

$|4x - 3| > 15 - 4 = 11$

Split into two outside cases

$4x - 3 < -11$ or $4x - 3 > 11$

Solve each case

Left side

$4x - 3 < -11$

$x < -2$

Right side

$4x - 3 > 11$

$x > 3.5$

Write the outside intervals

$x < -2$ or $x > 3.5$

Number line: $(-\infty, -2) \cup (3.5, \infty)$

$x < -2$ or $x > 3.5$

Teacher answers & working

Modulus and Absolute Value · Inequalities

ALG-C00B-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

Solve: $|3x| < 9$

Answer:

$-3 < x < 3$

Working:

Convert to a compound inequality

$-9 < 3x < 9$

Solve both boundary equations

Lower boundary

$3x = -9$

$x = -3$

Upper boundary

$3x = 9$

$x = 3$

Write the values between the boundaries

$-3 < x < 3$

Number line: (-3, 3)

$-3 < x < 3$

Teacher answers & working

Modulus and Absolute Value · Inequalities

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E2

Solve: $|2x| - 3 > 5$

Answer:

$x < -4$ or $x > 4$

Working:

Isolate the absolute value

$|2x| > 5 + 3 = 8$

Split into two outside cases

$2x < -8$ or $2x > 8$

Solve each case

Left side

$2x < -8$

$x < -4$

Right side

$2x > 8$

$x > 4$

Write the outside intervals

$x < -4$ or $x > 4$

Number line: $(-\infty, -4) \cup (4, \infty)$

$x < -4$ or $x > 4$

Teacher answers & working

Modulus and Absolute Value · Inequalities

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E3

Solve: $|5x + 6| < 12$

Answer:

$-3.6 < x < 1.2$

Working:

Convert to a compound inequality

$-12 < 5x + 6 < 12$

Solve both boundary equations

Lower boundary

$5x + 6 = -12$

$x = -3.6$

Upper boundary

$5x + 6 = 12$

$x = 1.2$

Write the values between the boundaries

$-3.6 < x < 1.2$

Number line: (-3.6, 1.2)

$-3.6 < x < 1.2$